

24/8/26

main()

```
{
    int A, B, C;
    A = 15;
    B = 7;
    C = A + B;
    printf("Add");
    printf("%d", C);
}
```

① int A → Declaration → A is a variable containing int type
② A = 15 → Definition of value is value of. → %d

25/8/26

main()

```
{
    float n1, n2;
    float Sum;

```

n1	n2	Sum
7.1	9.9	

n1 = 7.1;

n2 = 9.9;

```
printf("Add");
printf("%f", Sum); }
```

printf("Sum of %f", n1, n2, Sum) X

printf("Sum of %f and %f is %f", n1, n2, Sum)

n1	n2	Sum
7.3	6.2	13.5

Sum of 7.3 and 6.2 is 13.5

Size of int = 2 Byte.
" " float = 4 Byte.

31/8/2021

main ()

```
{ float div;  
  int A, B, Sum, sub, mul; div;
```

A = 1;

B = 2;

Sum = A + B;

sub = B - A;

mul = A * B;

div = B / A;

printf("Sum of %d and %d is %d", A, B, Sum);

printf("Sub of %d and %d is %d", B, A, sub);

printf("mul. of %d and %d is %d", A, B, mul);

printf("div. of %d and %d is %f", B, A, div);

}

Table of 4:-

main ()

```
{ int A, n1, n2, n3, n4, n5, n6, n7, n8, n9, n10; x1, x2,  
  A = 4; n3, x4, x5, x6, x7, x8, x9, x10;
```

n1 = 1;

n2 = 2;

n3 = 3;

n4 = 4;

n5 = 5;

n6 = 6;

n7 = 7;

n8 = 8;

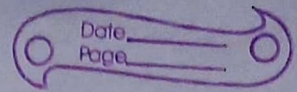
n9 = 9;

n10 = 10;

For optimization of Code- ① Storage

② Line of Code (LOC)

③ Execution (Output).



$x1 = A \times n1;$

$x2 = A \times n2;$

$x3 = A \times n3;$

$x4 = A \times n4;$

$x5 = A \times n5;$

$x6 = A \times n6;$

$x7 = A \times n7;$

$x8 = A \times n8;$

$x9 = A \times n9;$

$x10 = A \times n10;$

~~$\text{printf}(x1, x2, x3, x4, x5, x6, x7, x8, x9, x10);$~~

$\text{printf}(\text{"%.d, %.d, %.d, %.d, %.d, %.d, %.d, %.d, %.d, %.d", } x1, x2, x3, x4,$
 $x5, x6, x7, x8, x9, x10);$

}

↓

main()

{ int A, n;

A = 4;

n = A * 1;

printf("%.d", n);

n = A * 2;

printf("%.d", n);

n = A * 3;

printf("%.d", n);

n = A * 4;

printf("%.d", n);

n = A * 5;

printf("%.d", n);

```

n = A * 6;
printf("%d", n);
n = A * 7;
printf("%d", n);
n = A * 8;
printf("%d", n);
n = A * 9;
printf("%d", n);
n = A * 10;
printf("%d", n); }

```

1/9/26)

```

scanf("%d", &n1)
           ↓
        address of.

```

main()

```

{ int n1, n2, Sum;

```

```

    scanf("First Digit", &n1)
    scanf("Second Digit", &n2)

```

```

    printf("n1");
    scanf("%d", &n1);
    printf("n2");
    scanf("%d", &n2);
    Sum = n1 + n2;
    printf("Sum is %d", Sum);
}

```

$\gg, <, >, \leq, \geq$

2/9/26

$i = 2;$

T.B $\left[\begin{array}{l} \text{if } (i == 1) \\ \{ \\ \text{printf}("one"); \\ \} \\ \text{else} \\ \{ \\ \text{printf}("not one"); \\ \} \end{array} \right.$

$\text{if } (\underbrace{<\text{Condition}>}_{\text{replacable}})$

$\{$

True Block

$\}$

else {False Block}

7/9/26

lib. files $\rightarrow$ #include $\leftarrow$ stdio.h $\rightarrow$ std. input output

Pre Processor

$\text{void main}() \leftarrow \text{func}^n$

$\{ \text{int } n_1, n_2; \leftarrow \text{Variable declaration.}$

For	n_1	n_2	
	7	3	✓ Works
	3	7	✓ Works.
	5	5	X Don't Work

funcⁿ $\leftarrow$ $\text{printf}("Enter n_1 and n_2");$

formula specifics $\leftarrow$ $\text{scanf}("%d \%d", \&n_1, \&n_2);$ $\rightarrow$ add of

$\text{if } (n_1 > n_2)$

$\{ \text{printf}("n1 is greater");$

$\}$

else

$\{ \text{printf}("n2 is greater");$

$\}$

$\}$

```

I) if (n1 == n2)
    { printf("n1 and n2 are equal")
    }
    if (n1 > n2)
    { printf("n1 is greater than n2")
    }
    if (n2 > n1)
    { printf("n2 is greater than n1")
    }

```

for n1 n2
 7 3 ✓
 3 7 ✓
 5 5 ✓ } work.

```

II) if (n1 == n2)
    {
        // eq.
    }
    else
    {
        if (n1 > n2)
        { n1
        }
        else { n2
        }
    }

```

~~Divisible by no. $x \neq 1$ if ($x \neq 0$)~~
~~{ printf("Divisible") }~~

A no. divisible by other no:-

```
{ a, x, i;
    printf("Enter the no.");
    scanf("%d", &a);
    printf("Enter the divisor");
    scanf("%d", &i);
    x = a%i;

    if (x == 0)
    { printf("Divisible");
    }
    else { printf("Not Divisible");
    }
    return 0;
}
```

Q/p 9/261 - Greatest of 3 no. -

```
#include <stdio.h>
```

```
int main()
```

```
{ int n1, n2, n3;
```

```
printf("Give 1st No. - ");
```

```
scanf("%d", &n1);
```

```
printf("Give 2nd No. - ");
```

```
scanf("%d", &n2);
```

```
printf("Give 3rd No. - ");
```

```
scanf("%d", &n3);
```

```
if (n1 > n2)
```

```
{ if (n1 > n3)
```

```
{ printf("1st is greater");
```

```
}
```

```
if (n2 > n1)
```

```
{ if (n2 > n3)
```

```
{ printf("2nd is greater");
```

```
}
```

```
if (n3 > n1)
```

```
{ if (n3 > n2)
```

```
{ printf("3rd is greater");
```

```
}
```

```
return 0;
```

```
}
```

n1 n2 n3

2 5 8

8 2 5

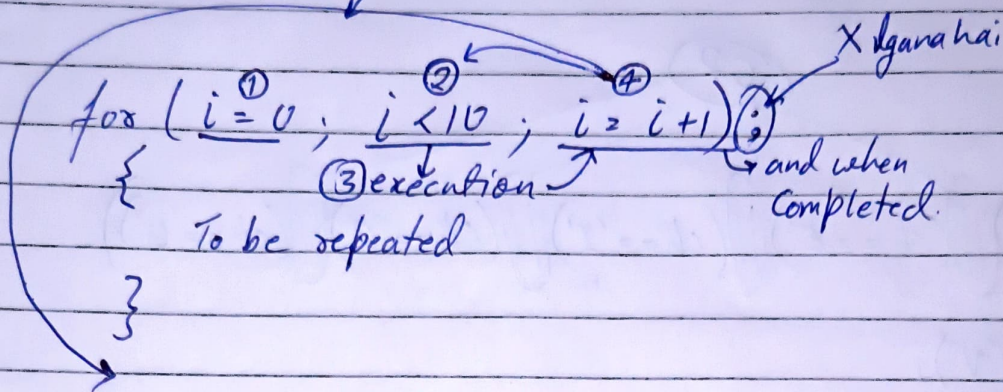
5 8 2

```
if (n1 > 2 && n1 > 5)
{ printf("1st");
}
```

```
if (n2 > n1 && n2 > n3)
{ printf("2nd");
}
```

```
if (n3 > n1 && n3 > n2)
{ printf("3rd");
}
```

14/9/26 - Loops - for while Do while.



⇒ Escape Sequence → like \n.

15/9/26 -

```

for (i = 1 ; i < 16 ; i = i + 1)
{
    printf("%d", i);
}

```

Switch-Case

```

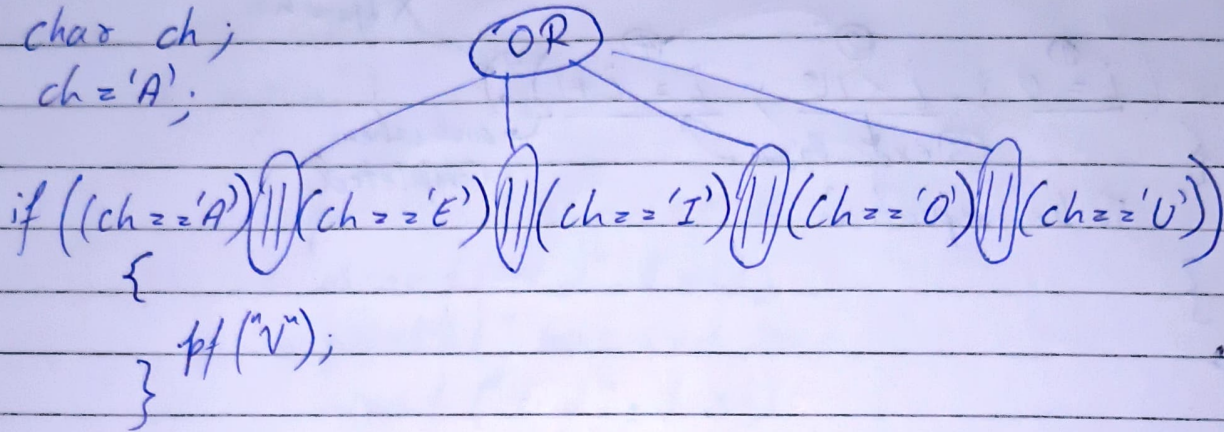
Switch (x)
{
    case 0...50
    {
        // ...
        break;
    }
    case 1
    {
        // ...
        break;
    }
}

```

Agar nhi lgaye, to case true uske case wale case hoga.

16/9/2021-

char ch;
ch = 'A';



How the code runs:-

$\begin{matrix} s & e \\ \boxed{2} & \boxed{15} \end{matrix}$

```

for (i = s; i <= e; i++)
{
    i = i + 1; i--;
    (i = i - 1)
    if (i % 2 == 0)
    {
        pf()
        cnt = cnt + 1; ← for counting.
    }
}
    
```

4 3 2 1

4 × 1000
3 × 100
2 × 10
1 × 1

1 2 3 4

1 × 1000
2 × 100
3 × 10
4 × 1